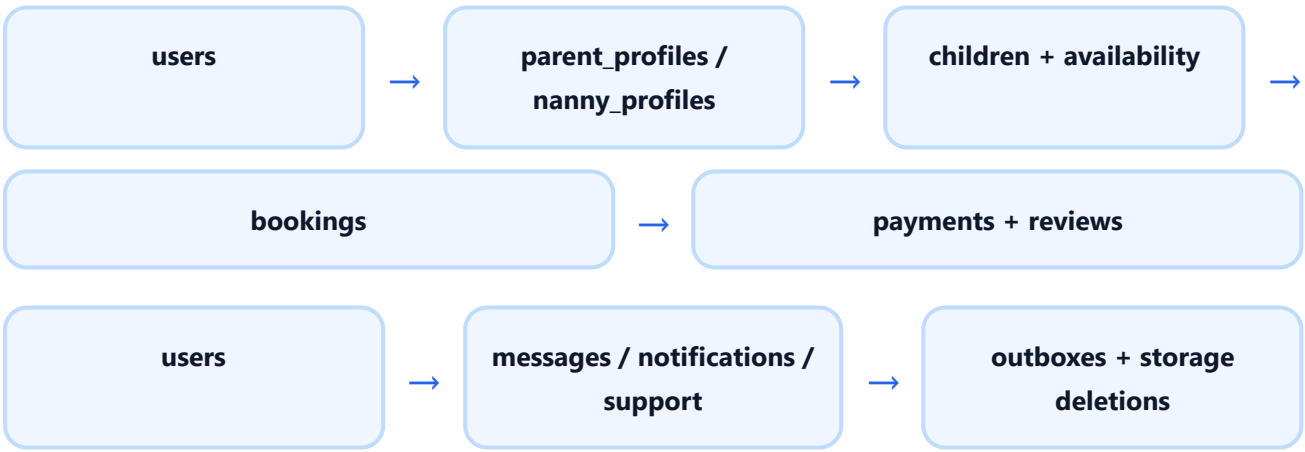
NANNY-APP · PRODUCT & ENGINEERING

Backend Schema & API Document

Data model, entity relationships, endpoints and contracts

Version 1.0 10 October 2026 Status: implementation baseline + launch requirements

1. Data model overview



2. Entity relationship reference

users
PK id
email UNIQUE
role, status
password_hash
email verification

nanny_profiles
PK/FK user_id
hourly_rate
verification_status
rating, bio, location

parent_profiles
PK/FK user_id
emergency contact
household data

children
PK id
FK parent_id
care details

bookings
PK id
FK parent_id, nanny_id
date_time, duration
status, PIN, dispute

payments
PK id
FK booking_id UNIQUE
amount, method
status, payout_status

chat_messages
sender_id → users
receiver_id → users
content, is_read

reviews
booking_id UNIQUE
reviewer_id, nanny_id
rating 1–5

Additional entities

`nanny_availability`, `nanny_portfolio`, `saved_nannies`, `api_tokens`, `password_resets`, `security_rate_limits`, `support_tickets`, `contact_messages`, `page_content`, `notifications`, `email_outbox`, `notification_outbox`, `storage_deletions` and `schema_migrations`.

3. Integrity and lifecycle rules

- InnoDB foreign keys link profile, child, booking, payment, message and review ownership.
- Booking creation/rescheduling locks the nanny row, validates every selected child and checks overlap in the same transaction.
- Booking and ledger state changes commit with notification events; failures roll back together.
- The recorded payment amount is the historical price source; later nanny-rate edits do not alter prior bookings.
- Migrations are ordered, checksummed and advisory-locked. Production uses the CLI runner and never destructive seed imports.

4. API endpoint catalogue

Domain	Endpoints	Auth
Authentication	`auth/register`, `login`, `logout`, `forgot`, `reset`, `verify-email`, `resend-verification`	Public except logout

User	`user/profile`, `change_password`, `upload_avatar`	Bearer
Nannies	`search`, `detail`, `save/saved`, `profile`, `portfolio`, `availability`, `earnings`, `reviews`	Bearer + role/ownership
Children	`list`, `create`, `update`, `delete`	Parent
Bookings	`list`, `detail`, `create`, `reschedule`, `accept`, `reject`, `cancel`, `check_in`, `check_out`, `confirm`, `dispute`	Bearer + participant/action
Messages	`conversations`, `thread`, `send`, `poll`, `mark_read`	Bearer + conversation participant
Notifications	`list`, `mark_read`, `mark_all_read`	Bearer owner
Payments/reviews	`payments/list`, `initialize`, `verify`; `reviews/create`, `list`	Bearer + booking role
Support/content/media	`support/create`, `my_tickets`; `content/page`; `media`	Mixed; media always authorized
Admin API	dashboard, users, bookings, payments, verification, support, refund and resolution routes	Admin bearer; Android login blocks admin

5. Data contracts

Response envelope

`{ "success": true|false, "message": string|null, "data": object|array|null }`. Errors use appropriate 400, 401, 403, 404, 405, 409, 429 or 503 status codes.

Booking representation

Includes IDs/reference, participant names, date/time, duration, address/location, notes/children details, status, agreed hourly rate/amount, authorized PIN visibility, check-in/out timestamps, dispute data, payment/payout status and created time.

Authentication

`Authorization: Bearer <opaque token>`. Only a token hash is stored. API clients must treat 401 as session loss and clear account-scoped local data.

Uploads

Multipart media is MIME/size checked and stored under generated private keys. Bytes are returned only after role/ownership checks. Android image selection is limited to 5 MiB before upload.

6. Data classification

Class	Examples	Controls
Restricted	Verification documents, child care details, reset/token material	Private storage, least privilege, no logs/public cache, encrypted backup.
Confidential	Messages, addresses, support and booking history	Participant/role authorization, HTTPS, retention policy.
Internal	Ledger and operational queue state	Admin/operator access, audit and monitoring.
Public	Approved nanny profile, ratings, published content	Moderation, safe output encoding and explicit publication.

Document suite